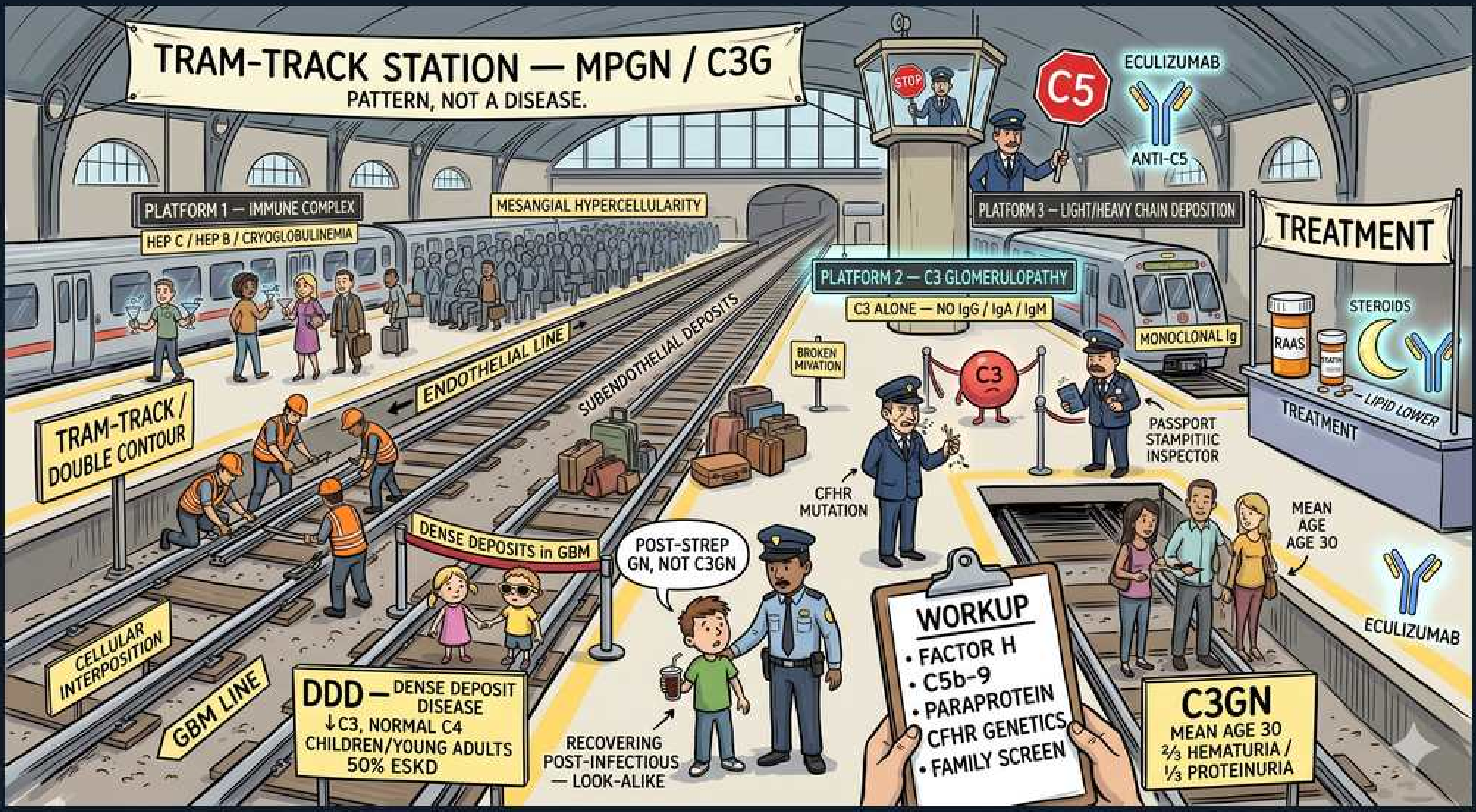
MPGN & C3 Glomerulopathy — Tram-Track Station



1. MPGN = pattern, not a disease

WHAT YOU SEE

Title banner 'Tram-track Station — MPGN/C3G, pattern not a disease'

WHAT IT ENCODES

Membranoproliferative GN is a histologic PATTERN (mesangial hypercellularity + capillary-wall double contours) with many causes. Modern classification splits by immunofluorescence into immune-complex-mediated vs complement (C3)-mediated.

BOARD TRAP

Treating 'MPGN' as one disease loses the point — always sub-classify by IF (Ig + complement vs C3-dominant).

2. Tram-track / double contour

WHAT YOU SEE

Tram-track / double contour panel, left

WHAT IT ENCODES

On light microscopy (PAS/silver) the GBM shows splitting — double contours/tram-tracking — from mesangial cell interposition and new basement membrane formation beneath subendothelial deposits.

BOARD TRAP

Double contours also occur in chronic TMA and transplant glomerulopathy — context (deposits on EM) confirms true MPGN pattern.

3. Subendothelial deposits

WHAT YOU SEE

Subendothelial deposits sign on platform

WHAT IT ENCODES

Immune-complex MPGN shows subendothelial (and mesangial) dense deposits on EM, with double contours from mesangial interposition. These activate the classical complement pathway.

BOARD TRAP

Subendothelial = MPGN/class IV lupus; SUBEPITHELIAL humps = post-strep; subepithelial spikes = membranous — deposit location is the key.

4. Immune-complex MPGN (Type I)

WHAT YOU SEE

Platform 1 — immune complex (Hep C, cryoglobulinemia)

WHAT IT ENCODES

Immune-complex MPGN: driven by chronic antigenemia — Hepatitis C (± mixed cryoglobulinemia), Hep B, chronic infection, SLE, monoclonal gammopathy. IF shows Ig PLUS C3. Classical pathway consumed → low C4 and C3.

BOARD TRAP

New 'MPGN' in an adult mandates a Hep C, cryoglobulin, and monoclonal protein (SPEP/serum free light chain) workup — missing the cause is the trap.

5. C3 glomerulopathy

WHAT YOU SEE

Platform 3 — C3 glomerulopathy (C3 alone, no Ig)

WHAT IT ENCODES

C3 glomerulopathy: IF shows C3 DOMINANT with little/no immunoglobulin, from dysregulation of the ALTERNATIVE complement pathway. Splits into C3GN and Dense Deposit Disease by EM. Low C3, normal C4.

BOARD TRAP

C3-dominant IF with scant Ig = alternative pathway (C3G), NOT immune-complex MPGN — this single IF finding redirects the entire workup.

6. Dense Deposit Disease (DDD)

WHAT YOU SEE

DDD panel — dense deposits in GBM

WHAT IT ENCODES

Dense Deposit Disease: ribbon-like, osmiophilic intramembranous dense deposits transforming the GBM on EM. Children/young adults, often with C3 nephritic factor (stabilizes C3 convertase) and drusen/lipodystrophy. ~50% reach ESKD.

BOARD TRAP

DDD deposits are INTRAMEMBRANOUS (within the GBM) — distinguish from C3GN where deposits are mesangial/subendothelial/subepithelial.

7. C3 nephritic factor

WHAT YOU SEE

C3 character / CFHR mutation area, center

WHAT IT ENCODES

C3 nephritic factor is an autoantibody that stabilizes the alternative-pathway C3 convertase (C3bBb), causing persistent C3 consumption. Found in DDD and some C3GN; also screen for Factor H/I deficiency and CFHR mutations.

BOARD TRAP

Persistently LOW C3 with NORMAL C4 points to alternative-pathway dysregulation (C3 nephritic factor / Factor H) — low C3 AND C4 suggests classical-pathway immune complex disease.

8. Low C3 (alternative pathway)

WHAT YOU SEE

C5 control tower / convertase imagery

WHAT IT ENCODES

C3 glomerulopathy persistently consumes C3 via the alternative pathway and terminal C5b-9. Serologic hallmark: low C3, normal C4. Eculizumab (anti-C5) targets terminal complement.

BOARD TRAP

A normal C3 does NOT exclude C3G (complement can normalize) — diagnosis rests on the C3-dominant IF, not the serum level.

9. Post-strep GN look-alike

WHAT YOU SEE

Officer 'post-strep GN, not C3GN' sign

WHAT IT ENCODES

Resolving post-infectious GN can transiently show C3-dominant staining and low C3, mimicking C3G. The discriminator is TIME: post-strep C3 normalizes within ~8 weeks; persistent hypocomplementemia >8–12 wk suggests underlying C3 glomerulopathy.

BOARD TRAP

Don't label a child after pharyngitis as C3G on one low C3 — recheck complement at 8 weeks; failure to normalize is what flips the diagnosis.

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10. Workup panel

WHAT YOU SEE

Workup card: Factor H, C5b-9, paraprotein, CFHR genetics, family screen

WHAT IT ENCODES

C3G workup: complement levels (C3, C4, CH50), alternative-pathway autoantibodies (C3NeF, anti-Factor H/B), soluble C5b-9, C3 nephritic factor, genetic testing (CFH, CFI, CFHR, C3, MCP), and serum/urine paraprotein — monoclonal gammopathy can drive C3G in older adults.

BOARD TRAP

In adults >50 with C3G, a paraprotein/MGRS is a frequent driver — skipping SPEP/free light chains misses a treatable cause.

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11. C3GN

WHAT YOU SEE

C3GN card: mean age 30, hematuria + proteinuria

WHAT IT ENCODES

C3 glomerulonephritis: C3-dominant deposits that are mesangial/subendothelial/subepithelial (NOT the dense ribbon of DDD). Presents with nephritic-range hematuria ± proteinuria, mean age ~30. Variable progression to ESKD.

BOARD TRAP

C3GN vs DDD is decided on EM deposit character (discrete vs ribbon-like intramembranous), not on IF — both are C3-dominant.

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12. Eculizumab / treatment

WHAT YOU SEE

Treatment shelf — eculizumab, steroids, anti-C5

WHAT IT ENCODES

C3G treatment: supportive RAAS blockade + lipid control; immunosuppression (steroids ± MMF) for active disease; eculizumab (anti-C5) for severe/rapidly progressive disease. Treat the cause in immune-complex MPGN (e.g., antivirals for Hep C).

BOARD TRAP

For Hep C-associated immune-complex MPGN the answer is treat the virus (DAAs) ± rituximab for cryoglobulinemia — not just immunosuppression.
